

The Full Set of Significant Single-Edit Swings

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Abstract—Companion artifact to “Rephrase Before You Act: Characterizing and Mitigating Language Sensitivity in Vision-Language-Action Models”. The paper charts one significant single-edit swing per task and edit category; this document charts every significant set the searches found, with exact p-values. The underlying per-phrase records, including the phrases themselves, are the two CSV files under **data/** in this repository.

I. THE FULL SET

A single-edit swing is a set of phrases for the same task that differ by a minimal edit and whose success rates differ significantly. For π_0 on SIMPLER Bridge, the search evaluated 105 candidate pairs and 66 survived a two-proportion z test ($z \geq 1.96$); grouped by task and edit, they form the 35 sets of Fig. 1 ($n = 72$ rollouts per phrase; basket task $n = 180$). For $\pi_{0.5}$ on LIBERO, the search evaluated 426 candidate pairs and 115 were significant ($|\Delta| \geq 18$ pp, $p < 0.05$), forming the 23 sets of Fig. 2 ($n = 50$ per phrase). Unlike the paper’s figures, nothing is curated away here: multi-edit sets and multiple sets of the same category on the same task are kept.

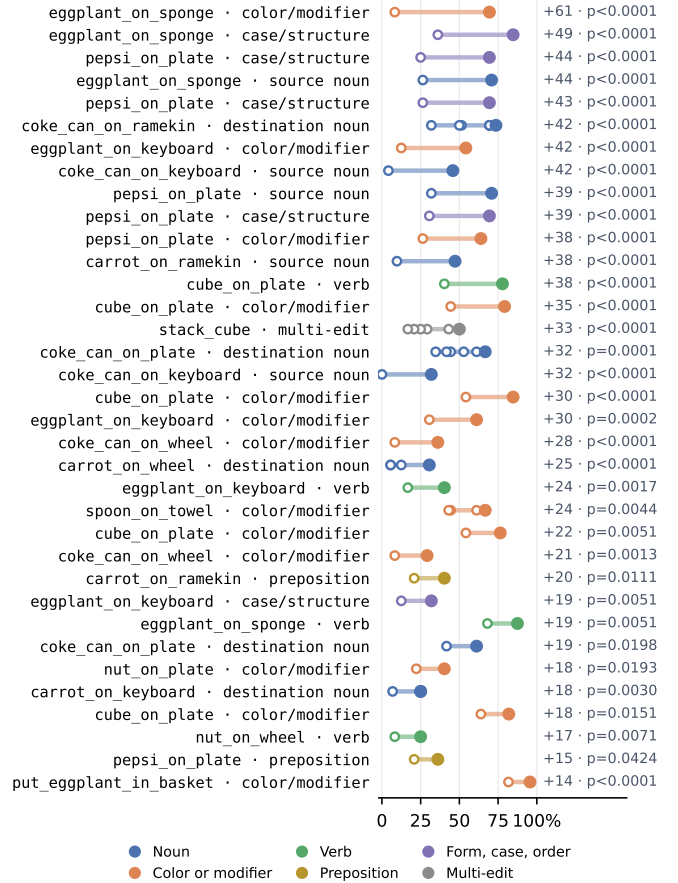


Fig. 1. All 35 significant swing sets for π_0 on SIMPLER Bridge, over 16 tasks, sorted by swing. Each row is one set; dots are its phrases’ success rates, the filled dot is the best phrasing, and the right column gives the swing in percentage points and the exact p of the set’s strongest pair under a two-proportion z test.

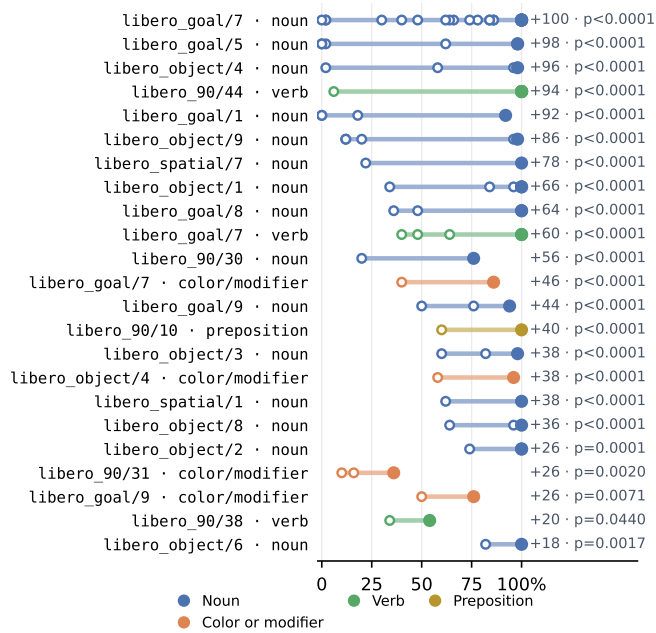


Fig. 2. All 23 significant swing sets for $\pi_{0.5}$ on LIBERO, over 19 tasks, formatted as Fig. 1. Of the 426 candidate pairs evaluated, 44% of in-finetune pairs showed a significant swing versus 5% of out-of-finetune pairs.